

$$\begin{aligned}
C_{\phi 1^2} \rightarrow & -\hat{m}_{H_u} + \frac{1}{16\pi^2} \left(\frac{1}{2} \sum_p g_1^2 \text{LF}_{1,0} [m_{\tilde{d}}^p] + \frac{1}{2} \sum_p g_1^2 \text{LF}_{1,0} [m_{\tilde{e}}^p] - \frac{1}{2} \sum_p g_1^2 \text{LF}_{1,0} [m_{\tilde{l}}^p] + \right. \\
& \left(3 \overline{y_u^{pr}} y_u^{pr} + \frac{1}{2} \sum_p g_1^2 \right) \text{LF}_{1,0} [m_{\tilde{q}}^p] - \sum_p g_1^2 \text{LF}_{1,0} [m_{\tilde{u}}^p] + 3 \overline{y_u^{pr}} y_u^{pr} \text{LF}_{1,0} [m_{\tilde{u}}^r] - \\
& g_1^2 \text{LF}_{1,1,-1} [m_1, \tilde{\mu}] - 3 g_2^2 \text{LF}_{1,1,-1} [m_2, \tilde{\mu}] + 3 \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} \text{LF}_{1,1,0} [m_{\tilde{d}}^r, m_{\tilde{q}}^p] + \\
& \left. \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} \text{LF}_{1,1,0} [m_{\tilde{e}}^r, m_{\tilde{l}}^p] + 3 \overline{a_u^{pr}} a_u^{pr} \text{LF}_{1,1,0} [m_{\tilde{q}}^p, m_{\tilde{u}}^r] \right)
\end{aligned}$$